

Vector embedding:

"It converts words or tokens into numerical vectors so that the neural network can work with language mathematically."

Self-attention:

"It allows each token to look at other tokens in the sequence and assign importance to them based on their context."

Feed-forward neural network:

"After attention, it further processes the information for each token using neural-network layers and nonlinear transformations."

1. Self-Attention — “Which words should I pay attention to?”

This is the **most important Transformer concept**.

Consider:

The animal didn't cross the road because it was tired.

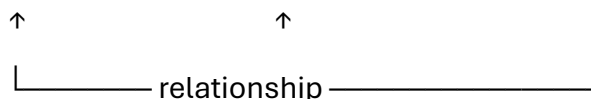
What does **"it"** refer to?

The model needs to understand the relationship between **"it"** and **"animal"**.

Self-attention allows each word/token to look at other tokens and determine **how important they are to understanding the current token**.

Conceptually:

The animal didn't cross the road because it was tired



The Transformer calculates attention scores to determine which tokens deserve more attention.

Self-attention allows every token to consider other tokens and determine which ones are important for understanding its meaning in context.

2. Vector Embeddings — “Convert words into numbers”

A Transformer cannot directly understand words like:

king, queen, car, happy

So the words/tokens are converted into vectors (lists of numbers).

For example, conceptually:

"king" → [0.21, -0.45, 0.78, ...]

"queen" → [0.19, -0.42, 0.81, ...]

Words with related meanings tend to have vectors that are mathematically closer to each other.

Think of it as:

Embedding = converting language into a numerical representation that the neural network can process.

3. Feed-Forward Neural Network — “Process what was learned”

After self-attention determines the relationships between words, the Transformer passes the resulting information through a Feed-Forward Neural Network (FFN).

Think of it as a small neural network that processes each token's representation and learns more complex patterns.

Simplified:

Input token representation

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Self-Attention

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Information about context

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Feed-Forward Network

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Refined representation

The FFN is essentially a couple of neural-network transformations with a nonlinear activation function in between.

You can think of it as:

Self-attention = understand relationships

Feed-forward network = transform/process that information

How the 3 fit together

The easiest way to remember the Transformer is:

Sentence

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Tokenization

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Vector Embeddings

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Self-Attention

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Feed-Forward Neural Network

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Better representation of the text

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Next Transformer layer

And a real Transformer has many such layers stacked together.